Rehabilitation Center Experience

Kennedy Krieger Institute | Neurobehavioral Unit (Inpatient)

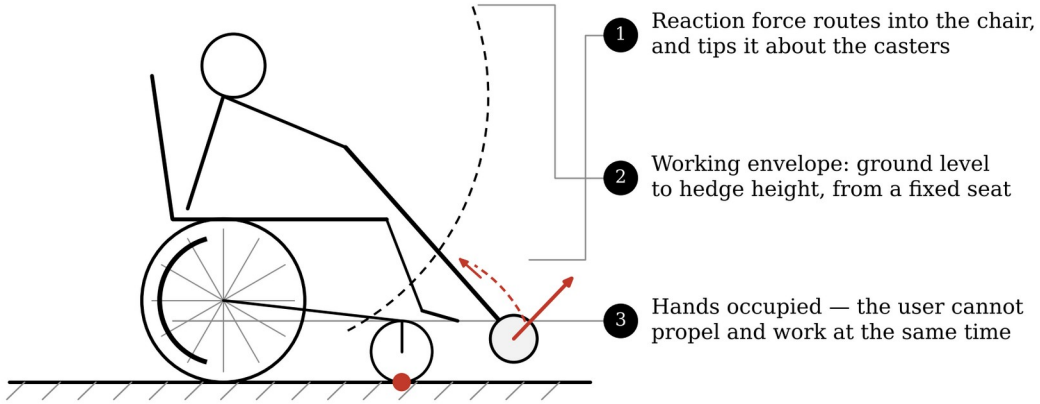
585.617 Rehabilitation Engineering | Module 10

Vance Van Volkenburgh | Visit: November 10, 2026

Project Topic

under one minute

- **Wheelchair-mounted interface for handheld powered yard tools** — trimmers, edgers, hedge trimmers, blowers.
- **Users:** manual wheelchair users with full upper-limb function, on their own residential yard.
- **Scope:** an interface and mounting problem. Not a new tool, not a new chair.
- **Three constraints are coupled**, and that coupling is the problem.
- No design has been selected. This visit was for input on the problem.



Manual wheelchair user operating a handheld powered yard tool on unimproved ground

The Center

Kennedy Krieger Institute, Baltimore, Maryland

- Pediatric **brain, spinal cord and musculoskeletal disorders**; treats individuals **up to age 21**.
- **25-bed inpatient rehabilitation unit**, accredited by CARF and the Joint Commission.
- Patients receive **several hours a day** of occupational, physical and speech therapy.
- **I visited the Neurobehavioral Unit (Inpatient)** — children and adolescents with developmental disabilities and severe problem behavior.
- Treatment is **applied behavior analysis**; admissions run weeks to months.



Patient Observation

what I was able to see

- I observed a **behavioral treatment session**, not a physical or occupational therapy session.
- What stood out was **how the session was structured**, rather than its clinical content:
  - Behavior was recorded **continuously and in real time**, not summarised afterward.
  - Staffing was high, and a second staff member was present **purely for safety**.
  - The environment was part of the treatment — surfaces, fixtures and equipment chosen for what they prevent.
- **Patient privacy limited what I could observe**, and I did not record or photograph anything on the unit.

What the Observation Changed

impact on my preliminary design

- **Measurement is part of the intervention.** Nothing was judged by impression. My PDS should carry **measurable acceptance criteria** — task time, number of attempts, setup time — not qualitative aims.
- **Staff presence is a design variable.** A task needing a second person is a different product. My equivalent question: can the user deploy and stow the tool **without a second pair of hands**?
- **Safety is designed into the environment before it is trained into the person.** I had treated stability as something the user manages by technique. The better framing is to make the unsafe state hard to reach.

Technology Overview

what is used, and how often

- **Used constantly, deliberately low-tech:** visual schedules, token boards, timers, picture cards, protective equipment, padded surfaces.
- **Used constantly, sophisticated:** real-time behavioural data capture on tablets, and video monitoring of sessions.
- **Used selectively, sophisticated:** augmentative and alternative communication devices, including speech-generating devices.
- **The most sophisticated element was methodological, not physical** — the functional analysis protocol itself. The hardware supporting it is simple on purpose.
- **Durability is the dominant hardware requirement.** Equipment must survive deliberate destruction, not just wear.

Relevance to My Project

nothing directly related, and why that matters

- **I saw no technology related to my project** — no wheelchair-mounted tooling, no outdoor task equipment, few manual wheelchair users.
- **That absence is structural.** The unit treats behaviour; it is not a mobility service, and its population is not mine.
- **But the constraints transfer, and they are harder than a rehab gym would impose:**
  - **Abuse tolerance** — against deliberate destruction, not incidental wear.
  - **Cleanability** — a requirement, not a preference.
  - **No ligature points, no entrapment, nothing readily detached or thrown.**
  - **Engagement as an engineering requirement** — if the user disengages, the device has failed.
- If my interface meets that bar, residential yard use is a far easier environment.

Care Provider Interviewed

Daniel Dobin, Clinical Assistant II

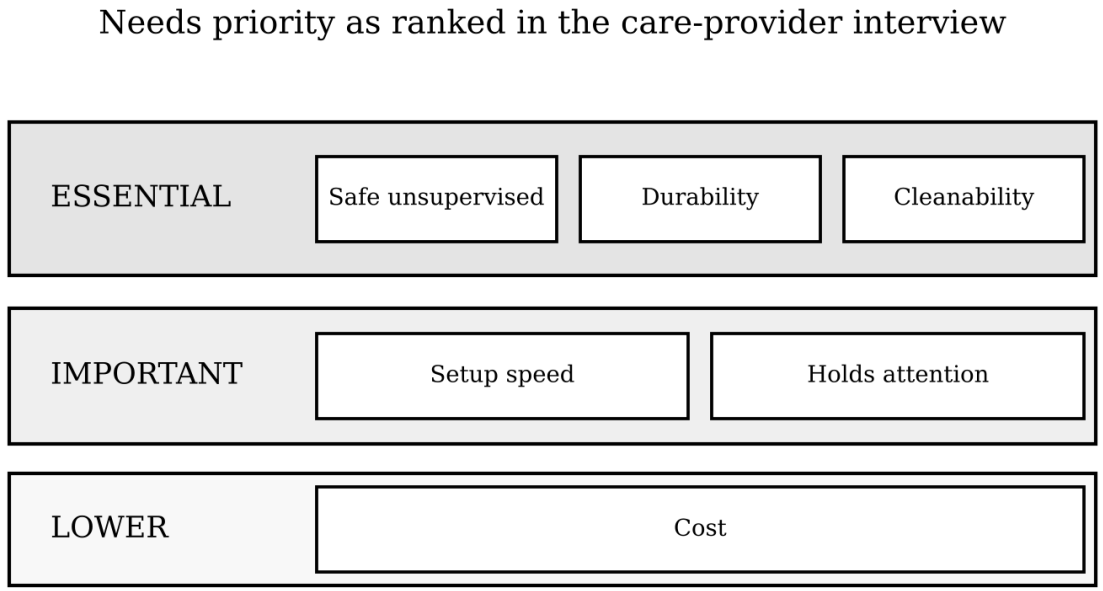
- **Clinical Assistant II, Neurobehavioral Unit (Inpatient)** — eighteen months on the unit, now **training incoming clinical assistants**.
- **Johns Hopkins University** undergraduate, 2024–2028.
- **Research Fellow, JHU Center for Suicide Prevention** — first author on systematic reviews in JAACAP and Neurotherapeutics, 2026.
- **Crisis Text Line** counsellor, two years.
- **Why this background mattered:** he sees equipment in daily use, trains the people who use it, and reads evidence critically.



Interview Themes (1 of 2)

constraints and priorities

- **Safety screening comes before function.** Ligature risk, pinch points, parts that detach or can be thrown — these are **pass/fail**, not trade-offs.
- **Cleanability sits with safety**, not with convenience.
- **Cost ranking lowest was the surprise.** I had assumed affordability was the binding constraint.



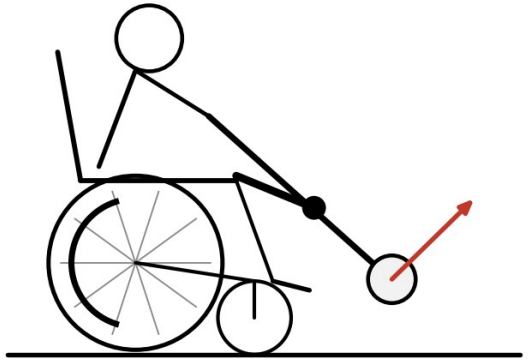
Cost ranking lowest was the surprise — it drives the Kano classification for the Preliminary Design Review

Interview Themes (2 of 2)

failure modes and solution directions

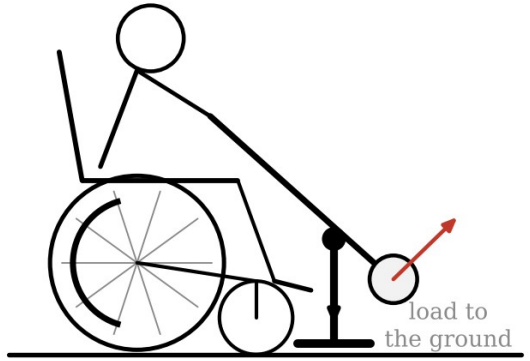
- **How devices actually fail:** breakage under deliberate force, lost parts, setups too slow to start, and patients disengaging — **mechanical failure was not the most common mode.**
- **On my three directions:** preference for **reducing the force at source**, since fewer attachments means fewer failure points and faster setup.

A — Carry it on the chair



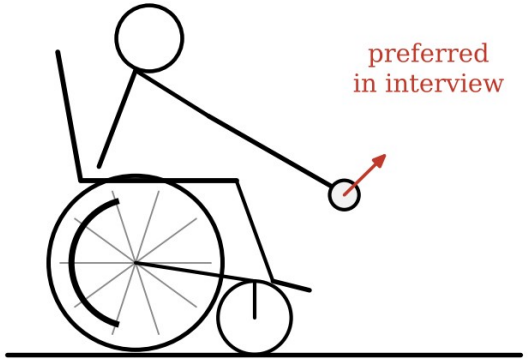
a mount carries the tool, so the reaction reacts through the chair

B — Route the force to the ground



a ground-bearing strut takes the load, leaving the chair unloaded

C — Reduce the force at source



lighter head, lower torque, nothing added to the chair

Impact on My Preliminary Design

what changes as a result

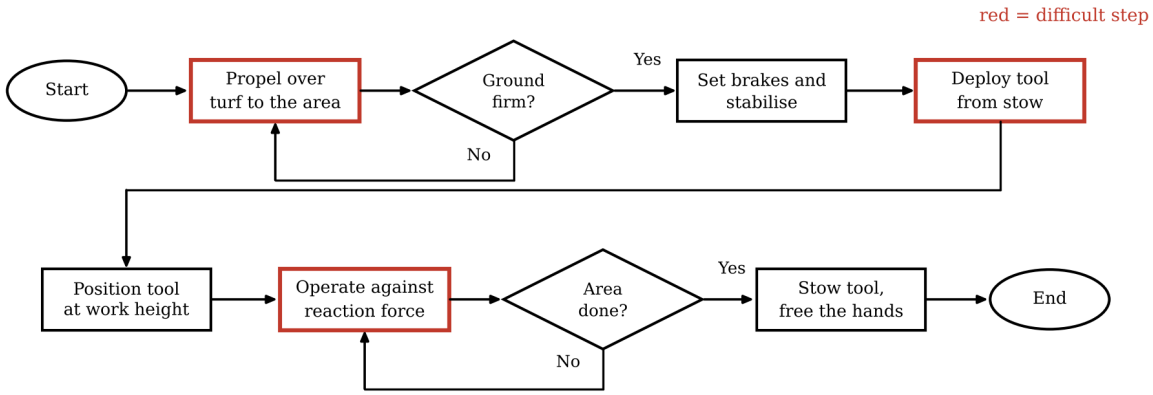
1. Quick-release becomes essential, not nice-to-have. Safety-driven, and the clearest single message of the visit.

2. Abuse tolerance and cleanability enter the PDS as requirements, not finishing details.

3. Abandonment becomes a design criterion. Setup time gets a targe

4. Acceptance criteria become measurable — attempts, elapsed time setup time.

5. Scope confirmed, source corrected. Quantitative inputs must come from an adult mobility clinician.



Where the design must act

What I Could Not Gather, and Next Steps

- **No client interview.** A pediatric inpatient behavioural unit; patient privacy and the clinical setting made a client conversation inappropriate. The user perspective is still missing.
- **No quantitative mobility data.** Tipping margins, propulsion penalties and reimbursement are outside a direct-care behavioural role.
- **Next steps:**
  - Follow up with a PT, OT or seating and mobility specialist for the quantitative inputs.
  - Interview adult manual wheelchair users — the population the project actually targets.
  - Carry the needs ranking into a Kano diagram for the Preliminary Design Review.
  - Use the safety and durability constraints as screening criteria in concept generation.